EXPERIMENT 2: WYE DELTA TRANSFORMATION

OBJECTIVE:

To understand on how to verify experimentally the principle of Wye-Delta and Delta-Wye transformation.

INTRODUCTION:

Wye-Delta Transformations

Situations often arise in circuit analysis when the resistors are neither in parallel nor in series. For example, consider the bridge circuit in Fig. 1. How do we combine resistors R1 through R6 when the resistors are neither in series nor in parallel? Many circuits of the type shown in Fig. 1 can be simplified by using three-terminal equivalent networks.

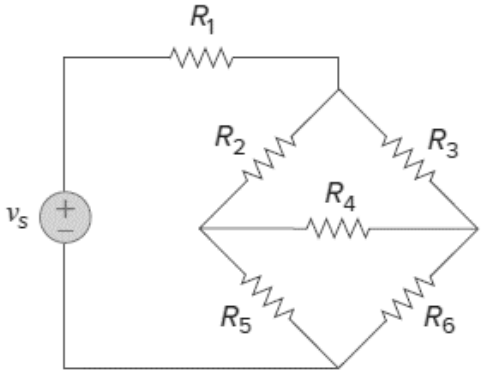


Figure 1: The Bridge network.

These are the wye (Y) or tee (T) network shown in Fig. 2 and the delta (Δ) or pi (Π) network shown in Fig.3. These networks occur by themselves or as part of a larger network.

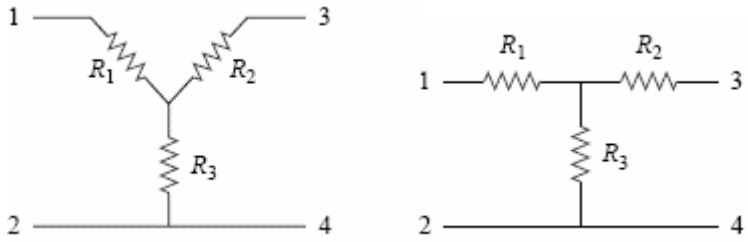


Figure 2: Two forms of the same network: (a) Y, (b) T.

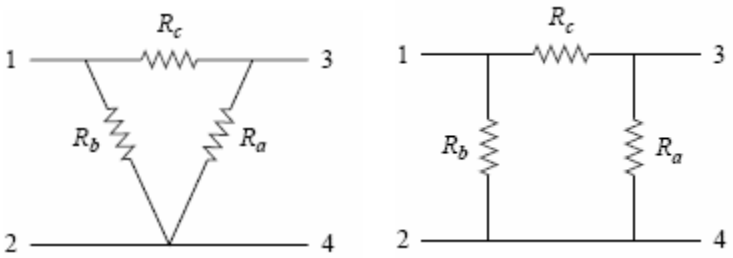


Figure 3: Two forms of the same network: (a) Δ, (b) Π.

Delta to Wye Conversion

Each resistor in the Y network is the product of the resistors in the two adjacent Δ branches, divided by the sum of the three Δ resistors.

Suppose it is more convenient to work with a wye network in a place where the circuit contains a delta configuration. We superimpose a wye network on the existing delta network and find the equivalent resistances in the wye network. To obtain the equivalent resistances in the wye network, we compare the two networks and make sure that the resistance between each pair of nodes in the Δ network is the same as the resistance between the same pair of nodes in the Y network. The following equations can be used for the transformation;

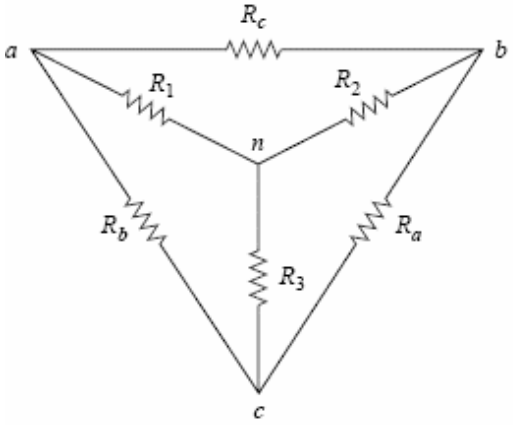


Figure 4: Superposition of Y and Δ networks as an aid in transforming one to the other.

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c} \quad (1)$$

$$R_2 = \frac{R_c R_a}{R_a + R_b + R_c} \quad (2)$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c} \quad (3)$$

Wye to Delta Conversion

Each resistor in the Δ network is the sum of all possible products of Y resistors taken two at a time, divided by the opposite Y resistor.

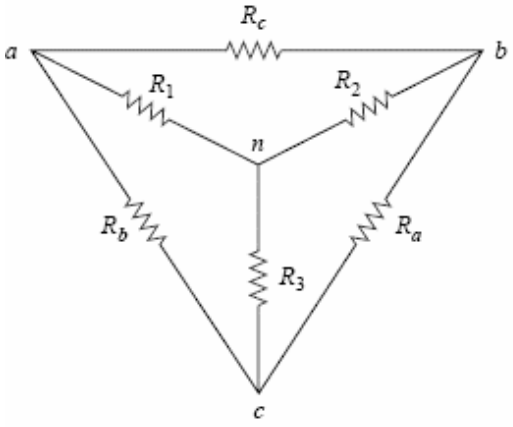


Figure 5: Superposition of Y and Δ networks as an aid in transforming one to the other

$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1} \quad (4)$$

$$R_b = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2} \quad (5)$$

$$R_c = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3} \quad (6)$$

EQUIPMENT AND COMPONENT

1. Digital Multimeter
2. Resistors

PROCEDURE

A. DELTA TO WYE CONVERSION

1. Construct the circuit in figure 6 and measure the resistance between a and b (R_{ab}^{1}). Write the result on table 1.
2. Then transform the 2k Ω, 4.7k Ω, 4.7k Ω part. Use delta to wye conversion, construct the new network with given resistances and measure the resistance again (R_{ab}^{2}). Then write the new value on table 1.
3. Also, draw the new network on the result section.

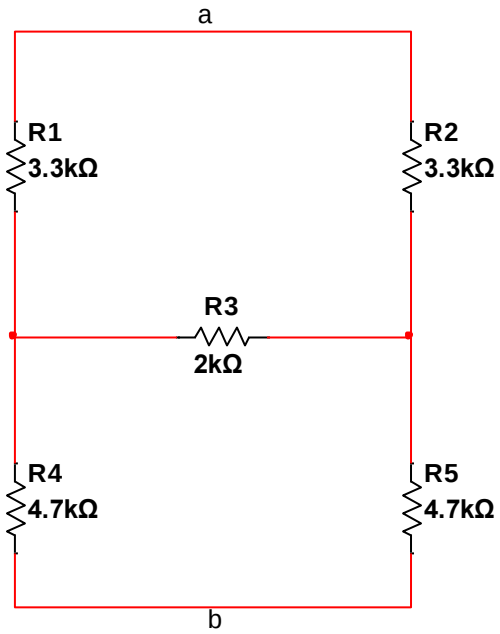


Figure 6

B: WYE TO DELTA CONVERSION

1. Construct the circuit in figure 7 and measure the resistance between a and b (R_{ab}^{1}). Write the result on table 2.
2. Then transform the 0.33K, 0.33K, 0.47K part. Use wye to delta conversion, construct the new network with given resistances and measure the resistance again (R_{ab}^{2}). Then write the new value on table 1.2.
3. Kindly draw the new network on the result section.

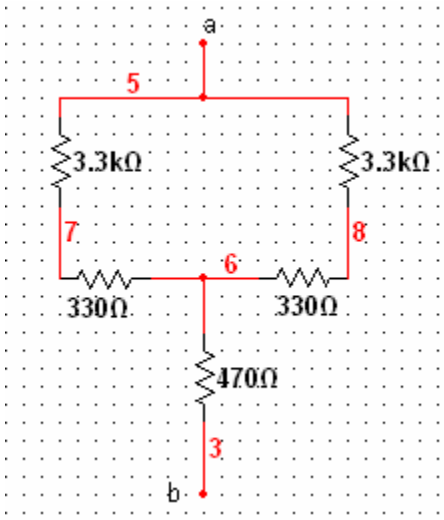


Figure 7

RESULT

PART A: DELTA TO WYE CONVERSION

Table 1

Resistor	Measured Value	Calculation value	Deviation %
R_{ab}^{1}			
R_{ab}^{2}			

Note: Please show all your working calculation and draw the new circuit networks.

PART B: WYE TO DELTA CONVERSION

Table 2

Resistor	Measured Value	Calculation value	Deviation %
R_{ab}^{1}			
R_{ab}^{2}			

Note: Please show all your working calculation and draw the new circuit networks.

DISCUSSION

1. Elaborate your result and findings in this experiment.
2. Show your working calculation for Table 1 and Table 2.
3. List the advantages of wye delta transformation.